Understanding China’s Industrial Policy Through Steel

Few industries reveal the logic of China's industrial policies as plainly as steel. It is too capital-intensive to develop without state coordination, too strategically important to leave entirely to markets, too politically entangled to rationalize without cost. For China, steel has been all of these things, and more. It has been a symbol of national sovereignty, an engine of the country's extraordinary growth, and now a test of whether that growth can be made sustainable. To understand how China governs its economy, one could follow the steel.

The industry's history in China spans nearly a century of turbulence. The early People's Republic inherited a shattered industrial base. Its most advanced steel infrastructure was seized by Japan in Manchuria, then destroyed by war, and rebuilt with Soviet assistance into the commanding heights of socialist construction. In 1958, Mao ordered a different approach: roughly 100 million people mobilized to smelt steel in backyard furnaces across the countryside. The campaign that produced millions of tonnes of metal too brittle to use, while harvests rotted in the fields. The Great Leap Forward's steel campaign is remembered as a catastrophe. It also foreshadowed the overcapacity issue in the steel industry.

The decades that followed traced a long arc away from the Great Leap Forward. The reform era introduced market mechanisms. Deng Xiaoping's dual-track pricing system allowed market forces to operate at the margins while preserving planned allocation, generating dynamism without surrendering control. Joining the WTO in 2001 turbocharged demand, and the 2008 stimulus package, which directed four trillion yuan heavily toward infrastructure, transformed China into the world's dominant steel producer. By 2014, China was producing nearly half the world's steel. The industry had become the backbone of an entire development model: feeding the construction of roads, railways, and cities; supplying the factories making ships, cars, and appliances; underpinning a three-decade expansion that lifted hundreds of millions out of poverty.

But that model created an overcapacity issue that China could not resolve from within. By 2015, China accounted for roughly half of global excess steel capacity. In 2016, Beijing attempted to solve this. The Supply-Side Structural Reform was an exercise in openly applied state power. Capacity targets were mandated by the central government, and inspection teams were dispatched to provinces. In total, 1.8 million redundancies were planned. It achieved real results but also ran into real limits, and overcapacity remains an issue to this day.

There appears to have been little done to address the overcapacity problem since 2016, and many countries have taken it upon themselves to protect their manufacturing. But tariffs and anti-dumping investigations have achieved little, with the Chinese finding ways to get around these measures. In order to design more efficient trade policies to target Chinese trade imbalances, one must understand Beijing’s industrial policy logic and the role traditional industries play in it, as well as their unique domestic conditions that restrain the central government’s policy choices.

Beijing recognizes overcapacity as a structural threat to industrial profitability, to price stability, and to its broader ambition of moving the economy toward higher-value production. The Chinese term *neijuan* — involution — entered official communication to describe what chronic overcapacity produces: a self-consuming race to the bottom in which firms compete relentlessly on price. Margins collapse, wages stagnate, and the resulting deflationary pressure suppresses domestic consumption that the Chinese economy desperately needs. To illustrate Beijing's own incentive to cut down capacity, we can look at the following official communications in the past two years. At a Politburo meeting on July 30, 2024, Xi called for industry to "prevent vicious involution competition" and for the "orderly exit of outdated and inefficient production capacities."[1] By the Central Economic Work Conference in December 2024, curbing involution had been elevated to the top economic priority for 2025.[2] Steel was named explicitly alongside cement and electric vehicles as a priority sector for anti-involution action — a signal that the overcapacity problem is now being reframed as a profitability and quality crisis requiring a different kind of intervention.[3] The 15th Five-Year Plan, adopted at the Fourth Plenum in October 2025 and finalized in March 2026, formalized this reframing: the NDRC stated that the raw materials industry must "deepen supply-side reform" through the 2026–2030 period, with "survival of the fittest" as the guiding principle, and committed to continued production controls on steel while prohibiting any new illegal capacity additions.[4] The plan performs two purposes: filtering out inefficient capacity and upgrading traditional industries through automation, digitalization, and green technologies. Taken together, these signals confirm that the center has not abandoned the overcapacity agenda after 2016. It has been embedded within a longer-horizon industrial policy that the green transition reflects.

The current push for green transition in the steel industry is from blast furnaces (BF/BOF) to electric arc furnaces (EAF). It is part of China's dual carbon commitments and ecological civilization agenda. But when examining the structural impact on industry, the EAF transition is using capital requirements, emission targets, and technology standards to let market forces eliminate inefficient mills. Where the 2016 reform asked provincial governments to accept the visible pain of losing tax revenue and unemployment, the second is designed so that the pain arrives through channels the central government does not directly control.

The contrast between these two reforms reflects a specific evolution in how Beijing manages the tensions of its political economy: between plan and market, between central goals and local interests, between domestic needs and global integration. The steel industry, sitting at the intersection of all three tensions, offers a clear lens into that evolution, and into the limits that no amount of policy sophistication has yet been able to dissolve.

I: The Problem

To understand why overcapacity issues persist in China, it is necessary to understand why the first restructuring attempt left so much undone and why the underlying problem proved resistant to one of the most direct exercises of central power.

The severity of the 2015 overcapacity issue can be demonstrated through a few figures. The industry was producing roughly 400 million tonnes more than domestic demand could absorb. Prices had fallen to fifteen-year lows, and return on assets across the sector had turned negative, with China's major steelmakers collectively losing RMB 53.1 billion between January and November of that year alone.[1] An estimated 51.4 percent of the industry consisted of firms that were losing money but continuing to operate, sustained by credit rollovers and local political support rather than any commercial logic.[2] The 2016 Supply-Side Structural Reform (SSSR) had to happen.

The SSSR's mechanism rested on three interlocking instruments. First, the central government directly mandated capacity closure targets down the party hierarchy, with 150 million tonnes to be cut by 2020 across major steel-producing regions. Second, state-owned banks were instructed to cease rolling over loans to firms on designated lists. This was implemented through direct instruction to the banking regulator rather than through local governments. It proved effective by bypassing the provincial hierarchy whose interests were misaligned with the center's goals. Finally, direct inspection teams were dispatched from Beijing to verify compliance. The combination produced the following results: steel prices rose more than 50 percent between 2015 and 2017 as marginal capacity exited and infrastructure-driven demand held firm.[3] The nominal target of 150 million tonnes of cuts was reported as achieved by 2018.[4]

The recovery in aggregate numbers obscured a more complicated reality on the ground. A significant portion of reported cuts involved capacity that had already been idle. The capacity replacement mechanism permitted new construction if equivalent old capacity was simultaneously retired, and provinces exploited it by inflating the capacity of aging mills before retirement and generating credits for larger new facilities. Despite nominal cuts, crude steel capacity began increasing again after 2017, returning to 2016 levels by 2019.[5] The consolidation targets fared no better: the top-ten steel producers' share of output (CR10), which the 13th Five-Year Plan had aimed to raise to 60 percent by 2020, moved only marginally from its 2015 baseline of 34 percent.[6]

The environmental record was starker still. The 2016 reform had no mechanism for addressing the carbon intensity of surviving production. The traditional BF/BOF route accounts for roughly 90 percent of Chinese steel output and produces approximately 2 tonnes of CO_2 per tonne of steel. When China announced its dual carbon targets in 2020, the steel sector presented perhaps the most intractable decarbonization challenge in the entire economy.[7]

What the 2016 reform left behind, then, was an industry that was somewhat leaner at the margins, considerably more profitable in the short term due to price recovery, but structurally not much changed from the one that had entered it.

It is against this background that the current push for EAF transition must be understood. The problem it addresses is not new. What has changed is the conditions under which restructuring must be attempted and the instruments available to do so. Those changes are substantial enough that a direct repeat of 2016 is neither politically feasible nor economically sensible. They also point, as the following sections argue, toward a reform logic that is, in some respects, more sophisticated than its predecessor — and, in others, more fragile.

II: Macroeconomic Conditions

The 2016 reform succeeded partly because the macroeconomic environment made it tolerable. China grew by 6.7 percent in 2016. Infrastructure stimulus sustained steel demand. Rising prices quickly restored profitability to the surviving mills, cushioning the economic blow to locals. The financial support from the central government, however inadequate in absolute terms, could be sold politically as a manageable transition cost against a backdrop of broader growth. The conditions of 2016, in short, created space for restructuring that the conditions of the mid-2020s do not.

The first and most structurally significant constraint is the risk of deflation. China's producer price index has now been in contraction for 40 consecutive months as of early 2026, with factory-gate prices falling 2.6 percent in 2025.[1] This is rooted in the collapse of real estate and declining global demand. In this environment, the political optics of the central government being seen to accelerate economic pain during a period of sustained price weakness would be deeply problematic. In 2016, the SSSR could be framed as a structural improvement that the price recovery validated. Today, no such validation is readily available.

The second constraint is employment, and it carries more than economic weight. China's urban youth unemployment rate reached a record 21.3 percent in June 2023, four times the overall unemployment rate, before the National Bureau of Statistics suspended publication of the series.[2] When the data resumed under a revised methodology excluding students, the rate stood at 16.9 percent in February 2025, and many analysts regard the true figure as substantially higher.[3] The broader labor market is one of structural stress: the twelve most labor-intensive industries shed 3.4 million jobs between 2019 and 2023,[4] and this year's cohort of university graduates, 12.7 million, is the largest in history.[5] In this environment, visible mass redundancies in a legacy industrial sector are a high risk to social stability.

The third constraint is local government fiscal stress, and it directly links steel restructuring to the fragility of China's national financial architecture. China's local governments have long relied on two revenue sources to fund public spending: land sale revenues and off-balance-sheet borrowing through local government financing vehicles (LGFVs). Both have been severely eroded. Land sale revenues contracted by 35 percent between 2021 and 2023 as the property market collapsed.[6] LGFV debt is estimated by the IMF at close to half of China's GDP. Many local governments are unable to service existing obligations without central help — by the end of 2022, twelve of China's thirty-one provinces were spending more than 100 percent of their monthly fiscal income on debt servicing.[7] In March 2024, Beijing took the unusual step of publicly instructing several provinces to cancel high-profile investment projects due to excessive debt levels.[8] In this context, the revenue loss that would follow forced mill closures — steel being a primary source of tax income in many provinces and LGFV collateral — could push already stressed local finances into crisis.

The fourth constraint is global trade. Chinese steel exports surged to a record 118 million tonnes in 2024, a 24 percent increase year-on-year and more than double the 2020 level. Weak domestic demand pushed surplus production into global markets.[9] In 2016, a similar export surge was absorbed internationally with significant but manageable friction. Today 81 anti-dumping investigations were initiated against Chinese steel by 21 countries in 2024 alone, a fivefold increase from 2023.[10] The EU has extended its steel safeguard measures and is implementing the Carbon Border Adjustment Mechanism (CBAM), which will apply fully to steel imports from 2026. The US Section 232 tariffs remain in place. Vietnam, South Korea, India, Brazil, and a growing list of emerging market producers have initiated their own investigations or imposed duties. The export option that helped absorb excess capacity in 2016 while domestic reform proceeded is closing — and it is closing precisely at the moment when domestic demand is at its weakest, and the pressure to export is therefore greatest.

Taken together, these four constraints define the political economy of the current moment with unusual precision. Every tool that worked in 2016 carries significantly higher political and economic risk. The central government needs a fundamental shift in the logic of intervention. Rather than acting directly on the industry, the center must design conditions under which the industry restructures itself. This is where the green transition comes in.

III: Reform Mechanism

The most consequential difference between the 2016 Reform and the current EAF transition lies in the implementation mechanisms. 2016 was an exercise in direct state power. Now the state designs conditions under which the industry restructures itself, and the proximate agent of change is the market rather than the ministry.

The 2016 mechanism produced issues like regulatory arbitrage, information manipulation, and the gradual reassertion of local interests once central pressure eased. The EAF transition deploys none of the instruments in the 2016 reform. Its primary mechanism is what might be called a capital-intensity filter. EAF technology requires substantial investment in furnace equipment, electrical infrastructure, and scrap procurement systems. This investment is accessible to large producers with state bank relationships, established credit ratings, and the organizational capacity to navigate green bond frameworks and transition loan programs; it is prohibitive for the small private mills that constitute the tail of China's fragmented industry. The central government does not need to order these mills to close. Their inability to finance the technological transition, combined with market competition, will accomplish the same result through market attrition.

This capital filter operates alongside a set of prohibitions that close the path for the old technology to grow, without mandating the closure of existing capacity. For the first time, provincial governments permitted no new BF/BOF projects in 2024, meaning all 7.1 million tonnes of newly approved capacity are EAF projects.[1] In August 2024, MIIT suspended the capacity replacement mechanism entirely, citing inadequate policy implementation and the need to craft updated frameworks for the 15th Five-Year Plan cycle.[2] This suspension closed the primary loophole exploited by provinces.

The green financing architecture reinforces this market selection logic. China's green bond market, in which SOEs accounted for roughly half of onshore issuances between 2019 and 2022, is structurally tilted toward large, state-connected producers.[3] Private enterprises, particularly those without government backing, face significantly higher barriers to green bond issuance and receive less favorable terms even when they qualify.[4] Green transition loans similarly flow to large firms with the organizational capacity to document emissions baselines and reduction trajectories. The industry has raised $3 billion via labeled bonds in 2024 and 42.5 billion yuan in transition loans as of early 2025,[5] but the distribution of these flows reinforces rather than corrects the scale advantage of the large SOEs. This achieves consolidation without coercing it.

The scrap supply chain serves as a third filter. EAF depends on reliable, competitively priced scrap steel. Large mills can establish procurement networks and maintain relationships with auto recyclers and demolition contractors to source scrap more efficiently. A smaller mill in inland Hebei cannot. It also favors mills near coastal regions, where scrap imports have low transportation costs.[6]

The result is a 'reform' whose effects are achieved through structural advantage rather than central order. This has a specific political value in the current macro environment. The central government retains what the 2016 reform never fully had — plausible distance from the employment consequences of restructuring.

However, green transition risks losing speed and certainty. The EAF target set by the government — 15 percent of crude steel production by 2025, and 20 percent by 2030 — has not been fully met. EAF production has remained stagnant at approximately 10 percent for over a decade, while utilization rates have fallen from 58.9 percent in 2021 to 48.6 percent in the first half of 2025, even as blast furnace utilization rose from 85.6 percent to 88.6 percent over the same period.[7] Global comparison makes this stagnation especially stark: the US produces 72 percent of its steel via EAF, India 58.8 percent, and the global average stands at roughly 30 percent.[8] China, despite having installed 151 million tonnes of EAF capacity as of early 2024, is running that capacity at rates that make the economics unfavorable.[9] The capital filter is working as a selection mechanism over the long run; it is not yet working as a decarbonization mechanism in the near term.

IV: Central-Provincial dynamic

It may be tempting to interpret the center-provincial dynamic in China as a simple story of central authority cascading down a hierarchy. The reality is more precise than this, and the precision matters for understanding why the 2016 reform failed where it did, and why the current EAF transition has been designed to avoid the specific forms of local resistance that defeated its predecessor.

The structural roots of that tension stemmed from the 1994 tax-sharing reform (*fenshuizhi*). Before 1994, local governments signed multi-year contracts with Beijing, agreeing to remit a fixed amount or a fixed percentage of revenue. Anything above the quota was retained for local expenditure. The reform centralized revenues sharply, with Beijing's share of total fiscal revenue rising from 22 percent in 1993 to 56 percent the following year. Meanwhile, the responsibilities of provinces remained unchanged.[1] Provincial governments accounted for approximately 79 percent of total government expenditure but only 47 percent of total government revenues.[2] To close this gap, local governments became dependent on industrial value-added tax and land sale revenues, and they needed good GDP figures on which their officials' promotion prospects depended.

The cadre evaluation system, formalized in the early 1990s, made GDP growth the dominant criterion for official advancement — a "hard target" in the system's language, one whose failure constitutes a veto over all other accomplishments.[3] Research confirms that prior to 2013, growth targets were directly linked to promotion probability for prefecture-level mayors, and that this produced systematic incentives to protect industrial output regardless of efficiency.[4] A county party secretary with a three-to-five year horizon before reassignment has every reason to support a local steel mill generating tax revenues and employment, and no reason to close it on behalf of a national efficiency objective. The result, compounded across hundreds of steel-producing localities, is what Lieberthal and Oksenberg call "fragmented authoritarianism": a state whose capacity is formidable at the level of policy design and severely constrained at the level of implementation, where local agents with misaligned incentives determine what actually happens.[5]

The 2016 Reform encountered this structural problem head-on and produced compliance failures. But it also produced two effective lessons whose logic the current reform has been designed around. The first was credit denial: by instructing the banking regulator directly, the center bypassed local governments entirely and imposed financial discipline on firms without asking local officials to act against their own interests. The second was environmental enforcement. The 2013–2017 Air Pollution Action Plan made air quality a mandatory hard target for officials in the Beijing-Tianjin-Hebei region. This put air quality in the same category as GDP growth. The failure to meet targets could block promotion.[5] Since steel mills in Hebei were responsible for 21 percent of total suspended particulate emissions and 28 percent of sulfur dioxide emissions in the region, officials who had previously protected mills found themselves facing career risk from tolerating them.[6] PM2.5 concentrations in the Beijing-Tianjin-Hebei region fell by nearly 25 percent between 2013 and 2017, and were the most significant pollution reduction achieved in the reform era.[7] The lesson is that the cadre evaluation system can be redesigned to become part of the solution.

The EAF transition applies this logic. Decarbonization is framed in terms of carbon intensity, a metric that, like air quality, is technically measurable and harder to falsify than capacity tonnage. It creates the conditions for enforcement through the cadre evaluation system rather than against it. Meanwhile, consolidation is the central government's long-run bet on dissolving the conflict of interests between central and provincial governments. The logic is that fewer, but larger and nationally-scaled producers are more tractable targets for compliance checks and less dependent on local government protection. Consolidation and decarbonization reinforce each other in a specific way: larger firms have the balance sheet to finance EAF conversion. The soft reform's two tracks — attrition of small mills through capital pressure, absorption of distressed capacity by large SOEs — are mutually reinforcing, and their combined effect on industry structure may prove more durable than either the 2016 reform's mandates or the EAF transition's market signals could achieve on their own.

V: Upstream and Downstream -- How the Strategic Context Changed

The EAF transition operates in a fundamentally different strategic context than in 2016. Its logic is simultaneously shaped by what Chinese steel depends on, what it supplies, and where its surplus goes. Each of these dimensions has changed substantially, and together they explain why the current restructuring is being pursued with a strategic coherence that 2016 did not quite have.

Upstream: Iron Ore and Raw Material Dependence

China imports approximately 80 percent of its iron ore, with Australia and Brazil accounting for the overwhelming majority of that supply.[1] In 2024, Australia alone accounted for 54.3 percent of global iron ore exports, while China consumed 73.4 percent of global supply — a bilateral dependency so deep that it has historically constrained China's diplomatic freedom of action.[2] In 2020, Beijing imposed trade restrictions on a range of Australian exports like barley, coal, wine, timber, and beef, in response to deteriorating bilateral relations. Iron ore was conspicuously absent from the list. China could sanction Australian agricultural and energy commodities; it could not sanction the input that feeds a billion tonnes of annual steel production without sanctioning itself.

Beijing has responded on two tracks. The first is geographic diversification through the Simandou project in Guinea, the world's largest untapped deposit of high-grade iron ore, valued at over $20 billion and nearly three decades in development. The first shipment of Simandou ore arrived at Majishan port in Zhejiang province in January 2026, carried by the Winning Youth vessel after a 46-day voyage from Guinea.[3] At full capacity, Simandou is projected to produce 120 million tonnes annually, approximately 10 percent of China's current import volume.[4] China Baowu, whose majority shareholding in the Winning Consortium Simandou gives it 31 million tonnes annually at full capacity, is both the world's largest steelmaker and now a primary investor in one of its most strategically significant new ore sources.[5] Simandou's high-grade ore — averaging 65 percent iron content versus the Pilbara's declining 60.8 percent — also happens to align with the specific requirements of lower-emissions steelmaking, since higher-grade feedstock reduces the energy and processing intensity of iron reduction.[6]

The second and more structurally transformative track is the EAF transition itself. China consumed 214 million tonnes of scrap in 2024, with domestic supply projected to rise to over 300 mt by 2030 as China's urbanization-era infrastructure reaches end-of-life.[7] If China achieves a 30 percent EAF share, one plausible scenario for the mid-2030s is that annual iron ore imports could fall by several hundred million tonnes, significantly reducing external dependency. EAF also relies on electricity, supported by China's increasingly mature and extensive green energy grid. The EAF transition is thus simultaneously an environmental policy, an industrial consolidation mechanism, and a resource security strategy whose logic operates on a long horizon. Beijing has also established the China Mineral Resources Group with 20 billion yuan in capital specifically to centralize iron ore purchasing and convert China's dominant demand share into actual negotiating leverage, an effort to reduce market vulnerability through buyer coordination rather than supply substitution.[8]

Downstream: Real Estate Out, Green Industries In

The 2016 reform was conducted against a backdrop of relatively stable downstream demand. Real estate and construction were slowing but had not yet collapsed. The current demand architecture has fundamentally changed. At peak, residential construction alone absorbed 31 percent of all steel consumed domestically in China.[9] That demand has contracted, and no single replacement has yet been able to fill.

The strategy has been to accelerate the development of downstream sectors that can absorb both the volume and the changing product mix of Chinese steel output. Electric vehicles, renewable energy infrastructure, and grid expansion have all been promoted as outlets for surplus production. China's Trade-in Program, launched in March 2024, is designed in part to stimulate replacement demand for steel-intensive goods and in part to increase the domestic supply of steel scrap.[10] EVs, wind and solar installations, and power grid expansion currently account for perhaps 8 to 12 percent of domestic steel demand. However, the gap between that figure and the 31 percent that real estate construction provided at peak is not comparable in the near term.

What the EV transition does provide, beyond partial demand substitution, is a pull toward specific steel grades. EVs require high-strength, lightweight steels, grades that are technically demanding and better aligned with EAF production capabilities than the traditional route. Automakers, like BMW-Brilliance, FAW-Volkswagen, and domestic manufacturers such as Chery, have begun signing offtake agreements specifically for low-carbon steel with reduced emissions intensity.[11] Baosteel has committed to supplying steel with 30 percent lower carbon intensity to Chery from 2024, scaling to over 50 percent reduction from 2026.[12] These agreements create demand that only large mills can meet efficiently. The demand structure is being reshaped in a way that reinforces the supply-side selection — a degree of policy coherence that the 2016 reform, which had no analogous demand-side logic, never achieved.

Conclusion

The wave of tariffs, anti-dumping investigations, and safeguard measures that China's export surge has triggered across the world rests on a common assumption: that raising the cost of Chinese exports will force China to confront its overcapacity more directly, reduce production, and compete on terms that other producers can match. That assumption misreads the political economy this essay has described. The export surplus is not a strategic choice that trade measures can price out but a symptom of domestic constraints that tariffs cannot reach. A central government that cannot impose direct capacity cuts without triggering unemployment, local fiscal stress, and deflationary pressure will not do so because foreign governments have made its products less welcome in their markets.

This is not an argument against trade measures as such. The argument is that trade measures alone will fail because they address a symptom rather than the cause. If the 2016 reform could not force the Chinese state to override its own domestic political economy, it is difficult to see why foreign tariffs would succeed where Beijing itself could not.

The key question then is what can be done about the trade imbalance. Answering it requires correctly diagnosing what China's advantage actually is and where it is going. China's current dominance is built on cheap products from cheap labor and manufacturing scale accumulated over decades. But Beijing knows this foundation is eroding. Wages are rising, demographic decline is shrinking the labor force, and the political cost of sustaining employment in inefficient industries is growing. The green transition and smart manufacturing agenda described in this essay is China's bet on retaining cost competitiveness. Automation makes machines cheaper than people everywhere, but China gets to deploy machines on top of an already low-cost ecosystem — abundant and increasingly clean energy, dense supplier networks, mature logistics infrastructure — that other countries cannot replicate quickly. The absolute cost floor China reaches through automation is therefore lower than what countries automating from a more expensive base can achieve. And because no other country has a policy cycle long enough to build that ecosystem, the gap does not close on its own. China is not trying to remain the world's cheapest manufacturer forever based on labor. It is trying to remain the world's cheapest manufacturer permanently by leveraging its infrastructure and integrating AI. Tariffs address the former. They have nothing to say about the latter.

Two responses follow. The first addresses scale and commercial interest simultaneously. No single Western country can match China's industrial volume alone, but allied manufacturing bases collectively can. Japan, Germany, South Korea, and Taiwan operate some of the world's most sophisticated industrial facilities, sitting on decades of process knowledge and supplier density that took as long to build as China's. Allies would also have incentives to cooperate with each other. For example, when China increases its green steel production capacity, the EU’s CBAM will be much less effective at reducing imports. What they lack is the AI and software infrastructure layer that converts manufacturing scale into the training data and deployment environment for the next generation of industrial systems. A coordinated US initiative to encourage deployment of American AI platforms across allied manufacturing supply chains would create an ecosystem that rivals China's in scale.

This is also where strategic and commercial interests converge. US AI companies face a pressing problem that is only partly technological: an investment bubble that has yet to demonstrate the concrete productivity gains that would justify the capital committed to it. Manufacturing is one of the applications to address that problem — first by providing the real-world environment in which American AI can demonstrate measurable productivity gains at scale, and then by converting that proof into a large and sophisticated commercial market for deployment. The returns flow in both directions: American AI gets the validation and revenue it needs; allies get the technology that makes their cost structures competitive against China's automating factories. A shared vulnerability becomes a shared asset, and the case for the alliance is made not in the language of security alone but in the language of commercial self-interest, which is more durable.

The second response addresses time. China's industrial strategy is designed to outlast Western electoral cycles — the green transition and anti-involution agenda in the 15th Five-Year Plan extends through 2030 and beyond. Democratic political systems find this horizon structurally difficult to match. The US response therefore requires institutional design that insulates industrial strategy from electoral volatility, not just bipartisan goodwill, which is fragile, but structural mechanisms that make reversal politically costly: treaty-level coordination with allies that embeds shared industrial commitments in international agreements, long-dated infrastructure mandates that outlast any single administration, and independent industrial policy institutions that operate across electoral cycles the way the Federal Reserve operates across them. China's bet is that Western political systems cannot hold a consistent industrial strategy long enough for it to matter. Proving that bet wrong is crucial.

Steel will remain the place to watch this play out. It is one of the industries where the contest between different models of industrial capitalism is being most directly joined. China is not waiting to be disciplined into competitiveness by foreign tariffs. It is quietly building new industrial competitiveness from which the next phase of its manufacturing dominance will be launched. The outcome of that contest is not yet determined. But the terms are set, and they were set, as so much else has been, in steel.

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